

ICLAB 2025 Fall Syllabus

Instructor:

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Time:

- Lecture: W 56 (13:30 ~ 15:30, Wednesday) @ ED415
- **Code Review: W AB (18:30 ~ 19:30, Wednesday) @ ED415 for Lab01~Lab04**

Prerequisites:

- Introduction to VLSI, Logic Design, Digital System Design, Computer Organization (Opt)

Course Objectives:

This course aims to convey the senior and graduated EE students techniques to design the VLSI chips using state-of-the-art CAD tools. In addition to learning CAD tools for performance-driven and cost effective IC designs, a top-down design flow and related environment will also be addressed. Upon completion of the course, the student will be able to design the integrated circuits and systems based on standard cell library as well as full-custom layout approaches. As such he/she will be able to work in a team of designers or stand alone.

Grading Policy

Weekly Lab Exercise x 12	(60%)
Bonus Lab	(3%)
Midterm Project	(8%)
Midterm Exam	(8%)
Online Test	(5%)
Final Project	(8%)
Final Exam	(8%)
Bonus (ADFP)	(?%)

Course Schedule:

Week	Date	Course Content	TA
1	09/3	00 、 Introduction + Environment Setting	王鈺翔
2	09/10	01 、 Cell Based Design Methodology + Verilog Combinational Circuit Programming	張祐榛
3	09/17	02 、 Finite State Machine + Verilog Sequential Circuit Programming	王英育
4	09/24	03 、 Verification & Simulation + Verilog Test Bench Programming	郭朝恩
5	10/01	04 、 Sequential Circuit Design II (STA + Pipeline) + Synopsys DesignWare IP	李崇碩
6	10/08	05 、 Memory & Coding Style (Memory Compiler)	蕭邦原
7	10/15	06 、 Synthesis Methodology (Design Compiler + IP Design) Midterm Project Announcement	陳致佑
8	10/22 10/24	No class : Study Days Midterm Exam + Online Test (Fri.)	
9	10/29	07 、 Timing: Cross Clock Domain + Synthesis Static Time Analysis	郭朝恩
10	11/05	08 、 Power: Low Power Design	王英育
11	11/12	09 、 System Verilog I (Design)	涂爾邑
12	11/19	10 、 System Verilog II (Verification)	涂爾邑
13	11/26	11 、 APR I : From RTL to GDSII	蕭邦原
14	12/03	12 、 APR II : IR-Drop Analysis + Final Project Announcement	陳致佑
15	12/10	13 、 Bonus (Formal Verification)	李崇碩
16	12/17	Final Exam	
17	12/24	TBD	TBD